NEURAL INFORMATION PROCESSING SYSTEMS
SAN DIEGO 2025

Rendering-Aware Reinforcement Learning for Vector Graphics Generation

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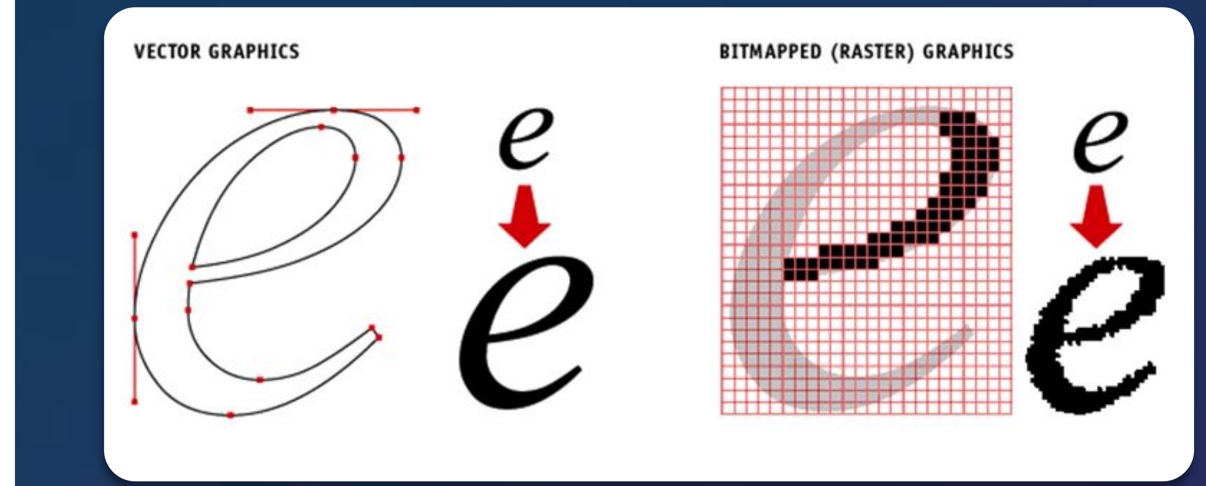
Project Website

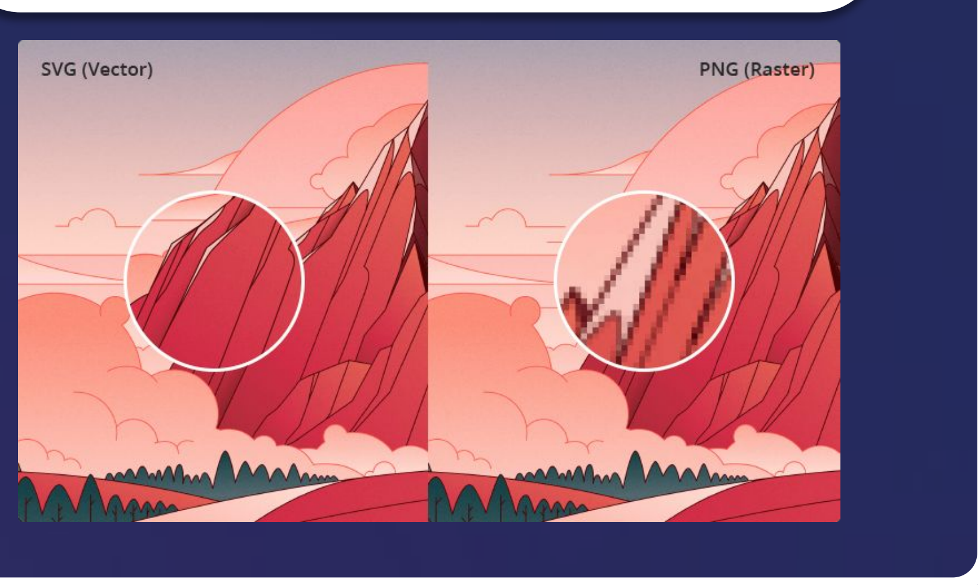
Connect on Linkedin

What are Scalable Vector Graphics?

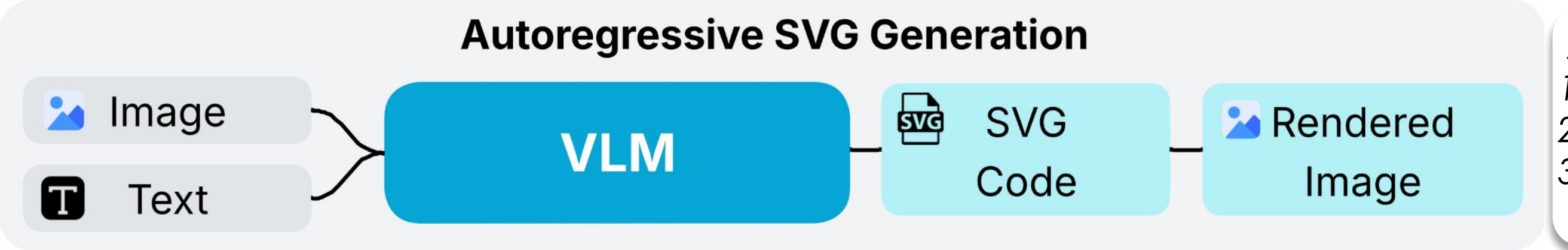


Key Vector Graphics Features:
1. *Mathematically Defined*
2. *Pre-Defined Primitives: Circles, Rectangles, Color Gradients*
3. *Scalable Image Resolution*
4. *Represented as SVG Code!*

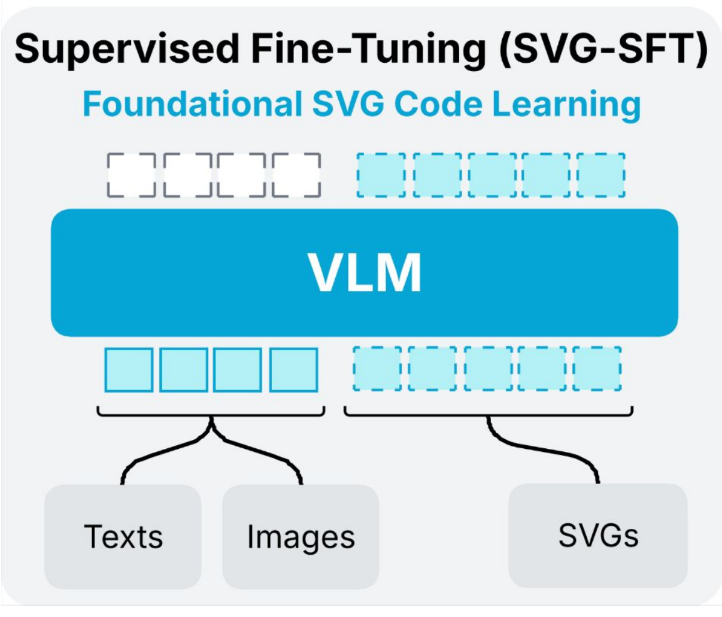
Advantages
1. *Sharp and precise images*
2. *Compact and lightweight*
3. *Editable: Just edit the code!*
4. *Used in Graphic Design, Logotypes, Website Development*

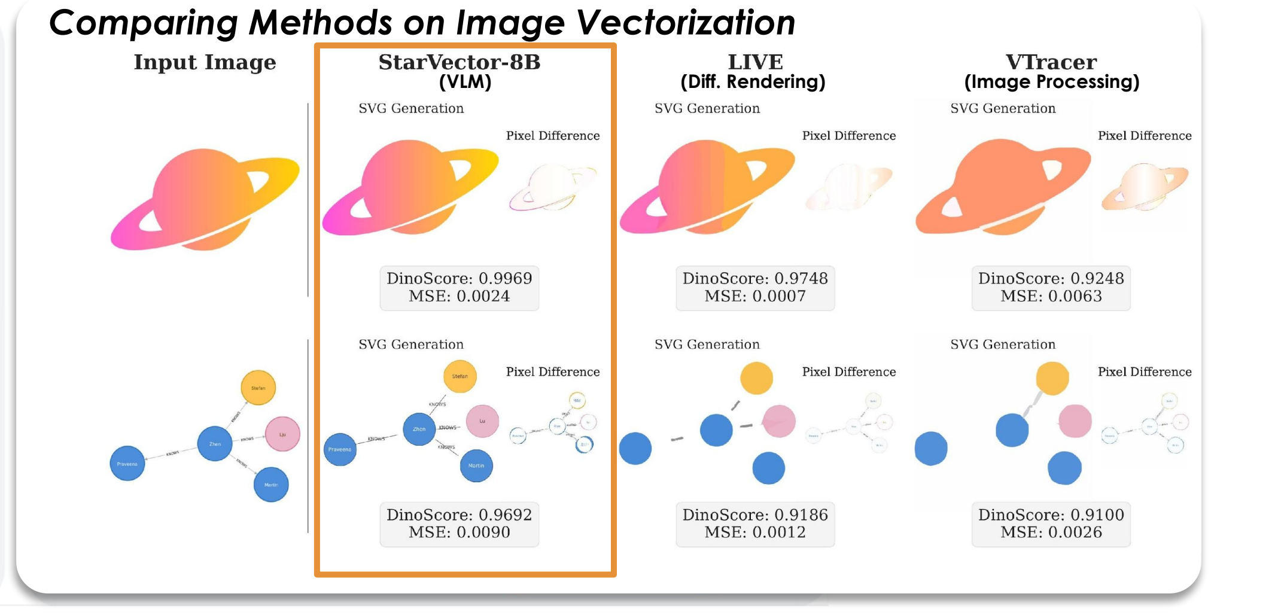


SVG Generation using Vision Language Models (VLMs)

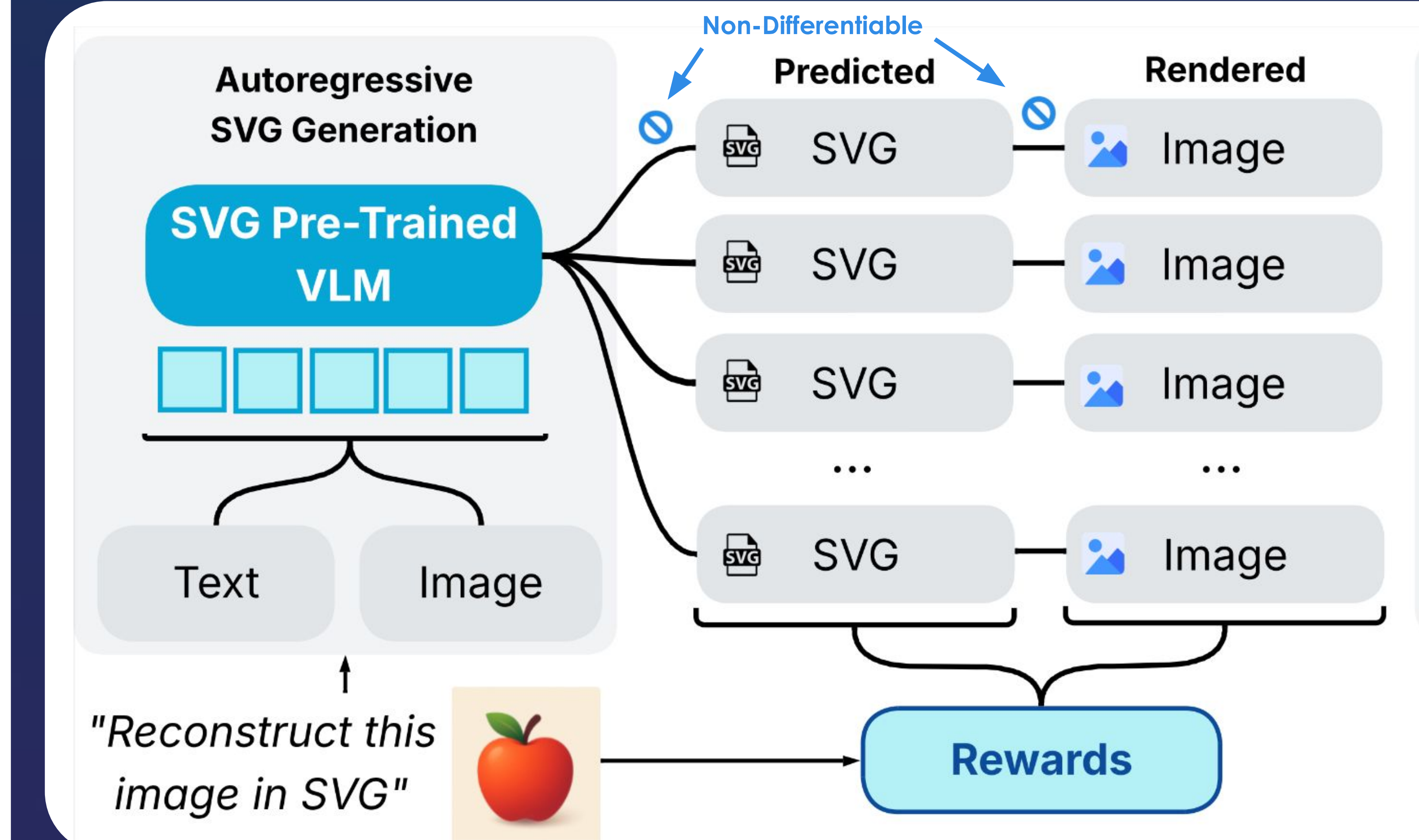


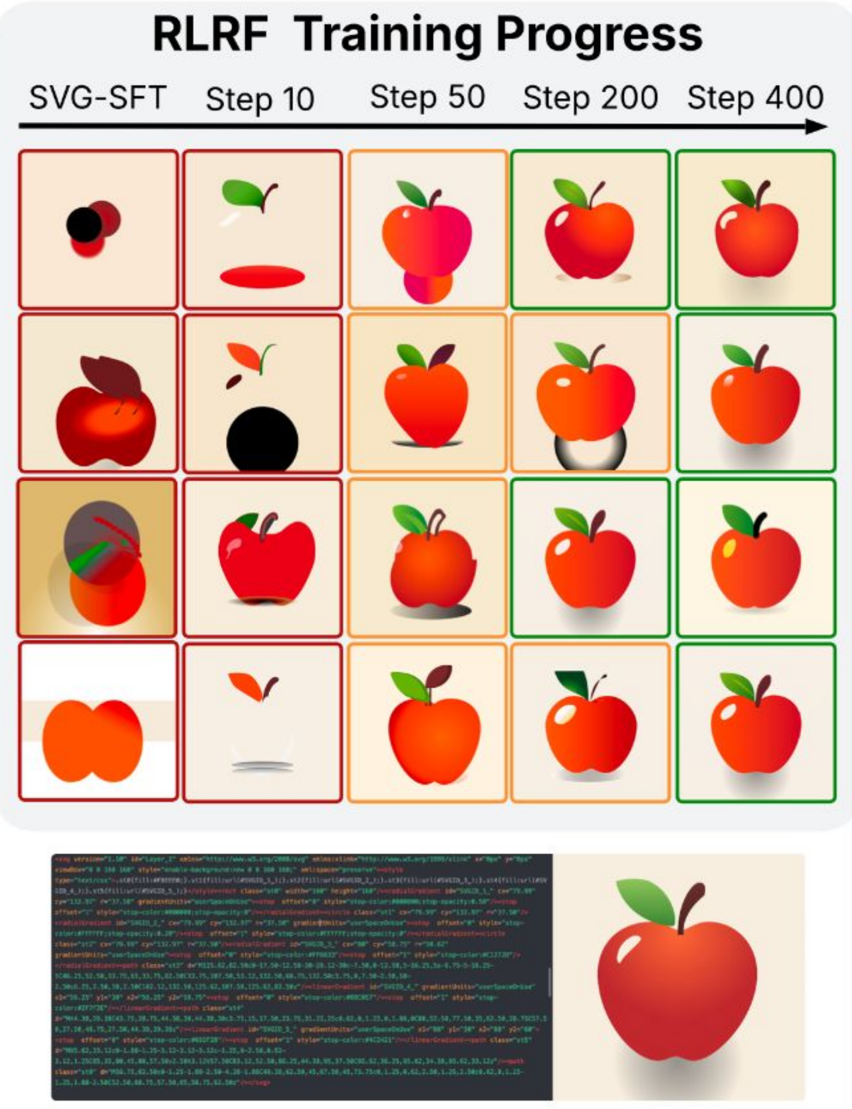
Advantages of VLMs
1. *Preserve SVG Primitives*
2. *Cleaner Vectorization*
3. *Text-Conditional Generation*


Supervised Fine-Tuning (SVG-SFT)
Foundational SVG Code Learning
VLM
Texts
Images
SVGs


Comparing Methods on Image Vectorization
Input Image
StarVector-8B (VLM)
LIVE (Diff. Rendering)
VTracer (Image Processing)
SVG Generation
Pixel Difference
DinoScore: 0.9969 MSE: 0.0024
DinoScore: 0.9748 MSE: 0.0007
DinoScore: 0.9248 MSE: 0.0063
DinoScore: 0.9692 MSE: 0.0090
DinoScore: 0.9186 MSE: 0.0012
DinoScore: 0.9100 MSE: 0.0026

RLRF (Reinforcement Learning from Rendering Feedback)





Reinforcement Learning with Rendering-Aware Rewards

Reconstruction

Pixel-level fidelity measured via **L2 distance** between the rendered SVG and the input image.

$R_{recon} = -||I - \hat{I}||^2$

Semantic Similarity

High-level perceptual alignment using **DreamSim** or **CLIP** embeddings.

$R_{sem} = \cos(\varphi(I), \varphi(\hat{I}))$

Code Efficiency

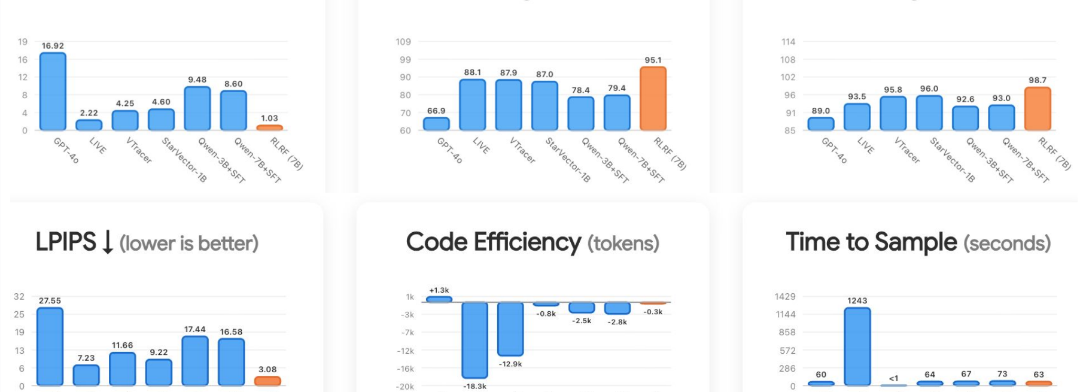
Encourages compact SVG code by penalizing **token length deviation** from ground truth.

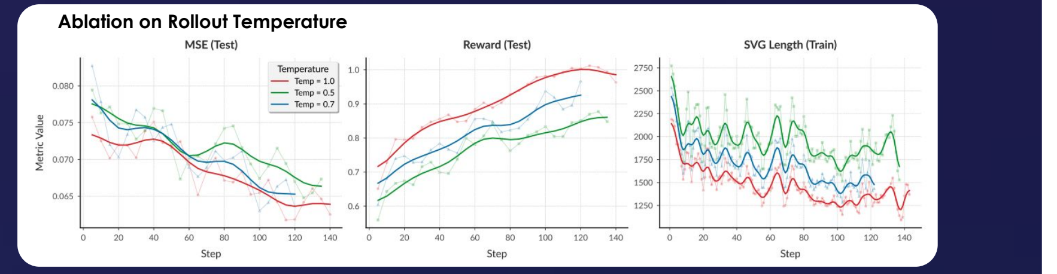
$R_{eff} = -|\mathrm{len}(\hat{y}) - \mathrm{len}(y)|$

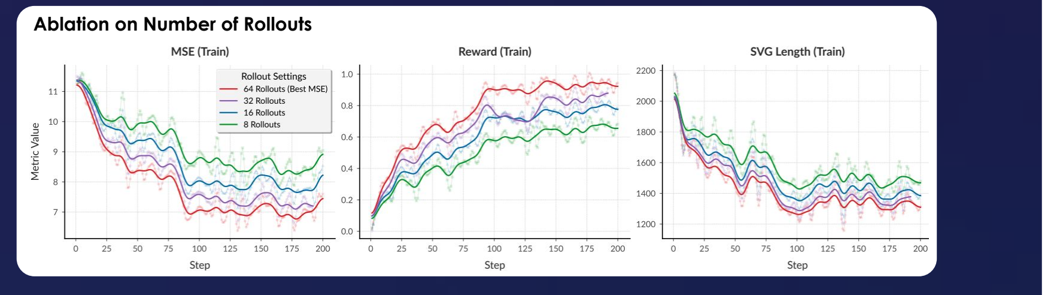
Experimental Results

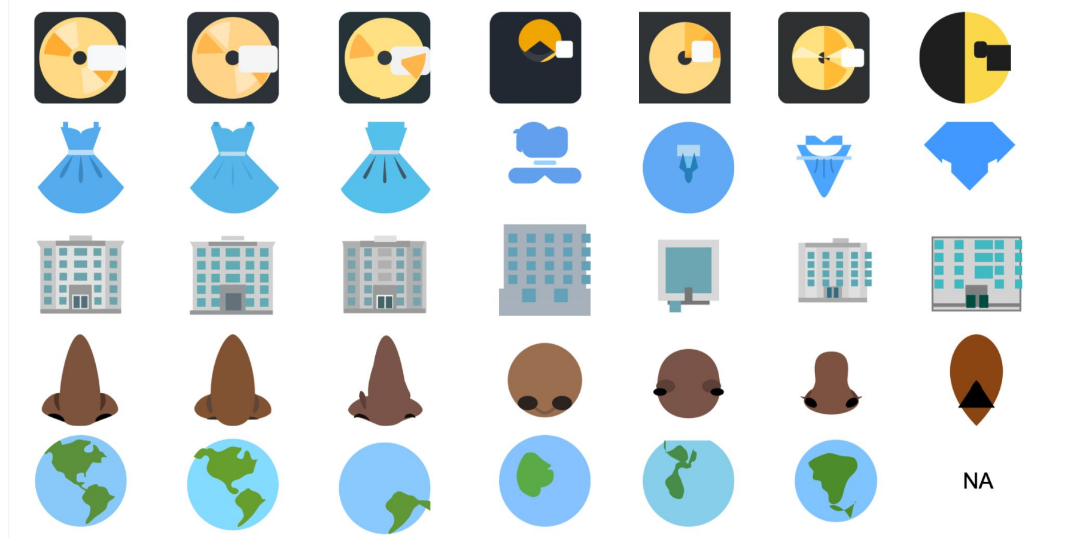
Performance on Image Vectorization

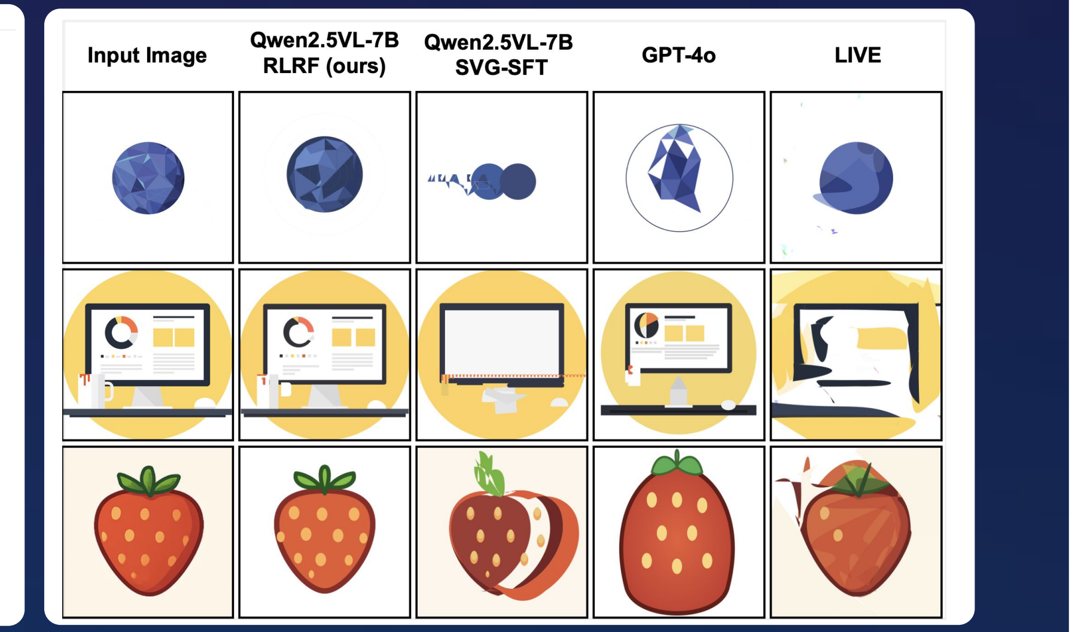
We perform **SFT** on **Qwen2.5VL-7B** with **1M Image-SVG** pairs (**SVG-SFT**) and then perform **RLRF post-training**





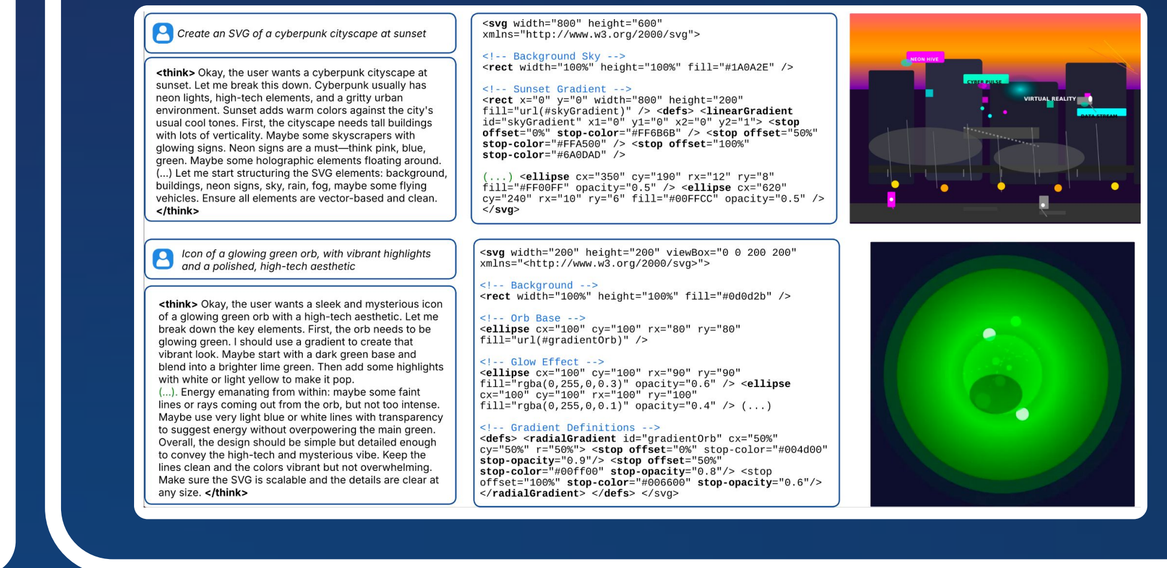






Performance on Text to SVG Generation

We perform **RLRF** on **Qwen3-7B** (not a VLM) with **Flickr30k Captions**.
We optimize for text-image alignment rewards, after rendering.


Create an SVG of a cyberpunk cityscape at sunset

<think> Okay, the user wants a cyberpunk cityscape at sunset. Let me break this down. Cyberpunk usually has neon lights, high-tech elements, and a gritty urban environment. Sunset adds warm colors against the city's usual cool tones. First, the cityscape needs tall buildings with lots of verticality. Maybe some skyscrapers with glowing signs. Neon signs are a must—think pink, blue, green. Maybe some holographic elements floating around. (...) Let me start structuring the SVG elements: background, buildings, neon signs, sky, rain, fog, maybe some flying vehicles. Ensure all elements are vector-based and clean. **</think>**

```
<svg width="800" height="600"
xmlns="http://www.w3.org/2000/svg">

<!-- Background Sky -->
<rect width="100%" height="100%" fill="#1A0A2E" />

<!-- Sunset Gradient -->
<rect x="0" y="0" width="800" height="200"
fill="url(#skyGradient)" /> <defs> <linearGradient
id="skyGradient" x1="0" y1="0" x2="0" y2="1"> <stop
offset="0%" stop-color="#FF6B6B" /> <stop offset="50%"
stop-color="#FFA500" /> <stop offset="100%"
stop-color="#6A0DAD" />

(...) <ellipse cx="350" cy="190" rx="12" ry="8"
fill="#FF00FF" opacity="0.5" /> <ellipse cx="620"
cy="240" rx="10" ry="6" fill="#00FFCC" opacity="0.5" />
</svg>
```

Icon of a glowing green orb, with vibrant highlights and a polished, high-tech aesthetic

<think> Okay, the user wants a sleek and mysterious icon of a glowing green orb with a high-tech aesthetic. Let me break down the key elements. First, the orb needs to be glowing green. I should use a gradient to create that vibrant look. Maybe start with a dark green base and blend into a brighter lime green. Then add some highlights with white or light yellow to make it pop. (...). Energy emanating from within: maybe some faint lines or rays coming out from the orb, but not too intense. Maybe use very light blue or white lines with transparency to suggest energy without overpowering the main green. Overall, the design should be simple but detailed enough to convey the high-tech and mysterious vibe. Keep the lines clean and the colors vibrant but not overwhelming. Make sure the SVG is scalable and the details are clear at any size. **</think>**

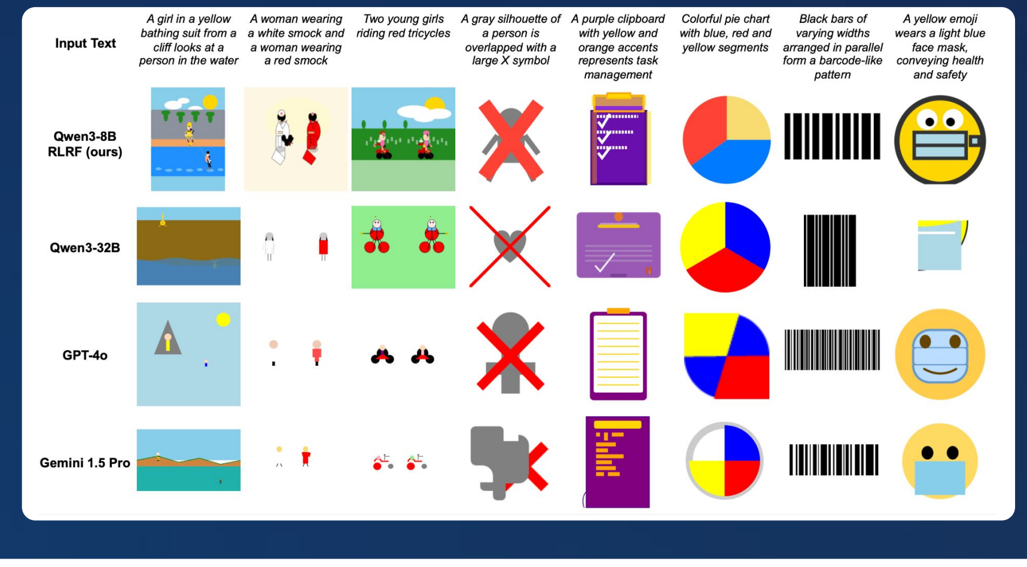
```
<svg width="200" height="200" viewBox="0 0 200 200"
xmlns="http://www.w3.org/2000/svg">

<!-- Background -->
<rect width="100%" height="100%" fill="#0d0d2b" />

<!-- Orb Base -->
<ellipse cx="100" cy="100" rx="80" ry="80"
fill="url(#gradientOrb)" />

<!-- Glow Effect -->
<ellipse cx="100" cy="100" rx="90" ry="90"
fill="rgba(0,255,0,0.3)" opacity="0.6" /> <ellipse
cx="100" cy="100" rx="100" ry="100"
fill="rgba(0,255,0,0.1)" opacity="0.4" /> (...)

<!-- Gradient Definitions -->
<defs> <radialGradient id="gradientOrb" cx="50%"
cy="50%" r="50%"> <stop offset="0%" stop-color="#004d00"
stop-opacity="0.9"/> <stop offset="50%"
stop-color="#00ff00" stop-opacity="0.8"/> <stop
offset="100%" stop-color="#006600" stop-opacity="0.6"/>
</radialGradient> </defs> </svg>
```


Input Text
A girl in a yellow bathing suit from a cliff looks at a person in the water
A woman wearing a white smock and a woman wearing a red smock
Two young girls riding red tricycles
A gray silhouette of a person is overlapped with a large X symbol
A purple clipboard with yellow and orange accents represents task management
Colorful pie chart with blue, red and yellow segments
Black bars of varying widths arranged in parallel form a barcode-like pattern
A yellow emoji wears a light blue face mask, conveying health and safety
Qwen3-8B RLRF (ours)
Qwen3-32B
GPT-4o
Gemini 1.5 Pro